Design of Enterprise IoT

Lecture 3

IoT Network

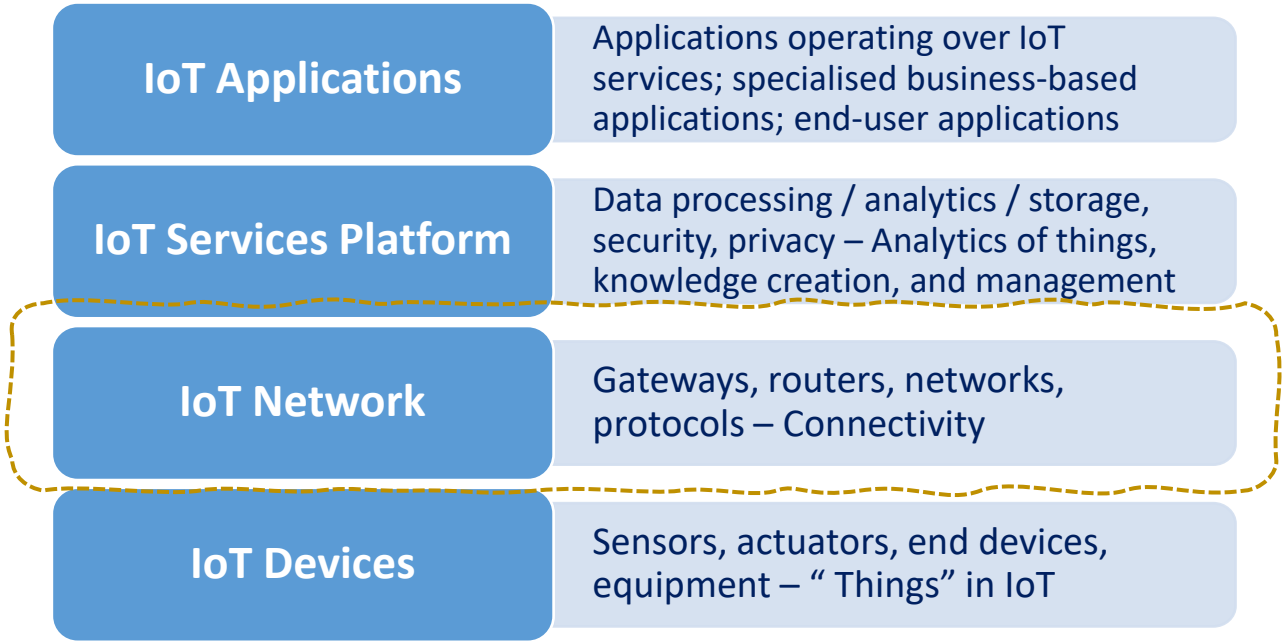


A/Prof Chun Ouyang

Recap: IoT Devices

- Sensing
  - Identify and collect key parameters for analysis, monitoring, and diagnosis
  - Convert analogue input to digital output
    - The signals from sensors must be communicated to other locations for aggregation and analysis
- Actuating
  - Address the need for controlling the “things” by performing action (i.e. sense and actuate)
  - Convert digital input to physical event/action
- Unique addressing/identification

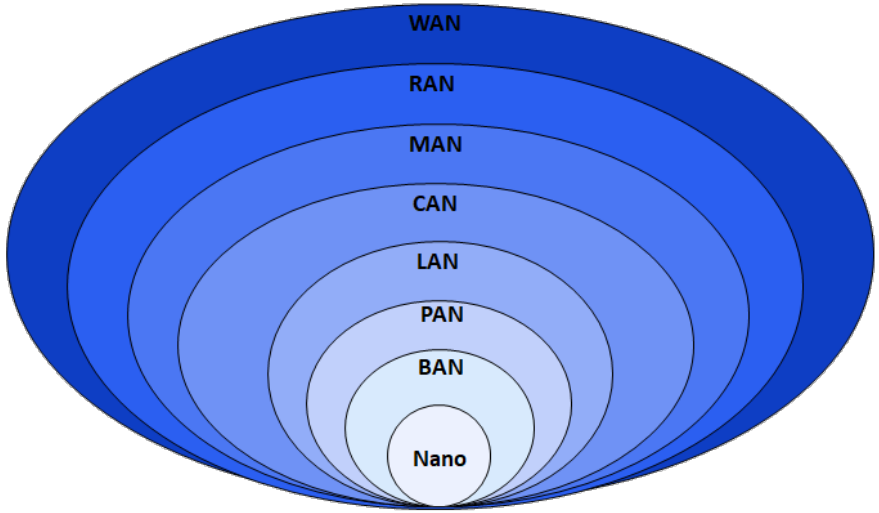
IoT Reference Framework – Network



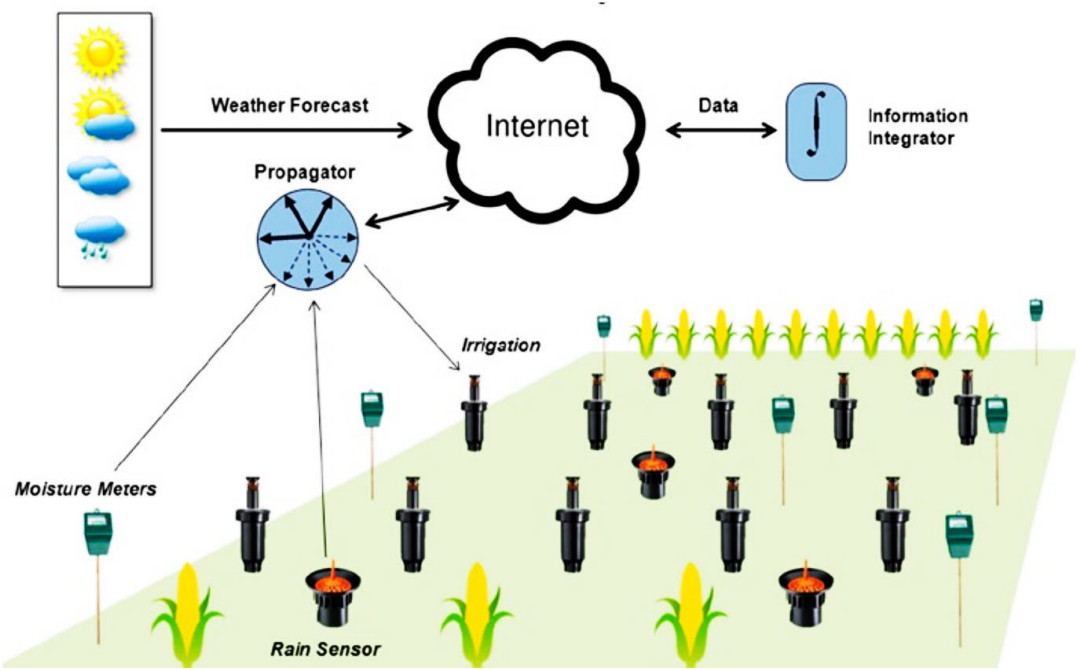
(Rayes and Salam, 2022)

Computer Network Types

Abbreviation	Full Name	Coverage / Meaning
Nano	**Nanonetwork**	Network at the nanoscale (molecular/nanoscale communication, e.g., within a few micrometers to millimeters)
BAN	**Body Area Network**	Network of wearable/implanted devices around or inside a human body (typically ~1–2 meters)
PAN	**Personal Area Network**	Network around an individual person (typically ~10 meters, e.g., Bluetooth)
LAN	**Local Area Network**	Network within a limited area such as a home, office, or building (typically up to ~1 km)
CAN	**Campus Area Network**	Network spanning a campus (e.g., university or corporate campus), larger than a LAN but smaller than a MAN
MAN	**Metropolitan Area Network**	Network spanning a city or metropolitan region (typically up to ~50 km)
RAN	**Radio Access Network**	The radio portion of a mobile/cellular network connecting user devices to the core network (not strictly a "size-based" category like the others)
WAN	**Wide Area Network**	Network spanning large geographic areas (countries, continents, the Internet)

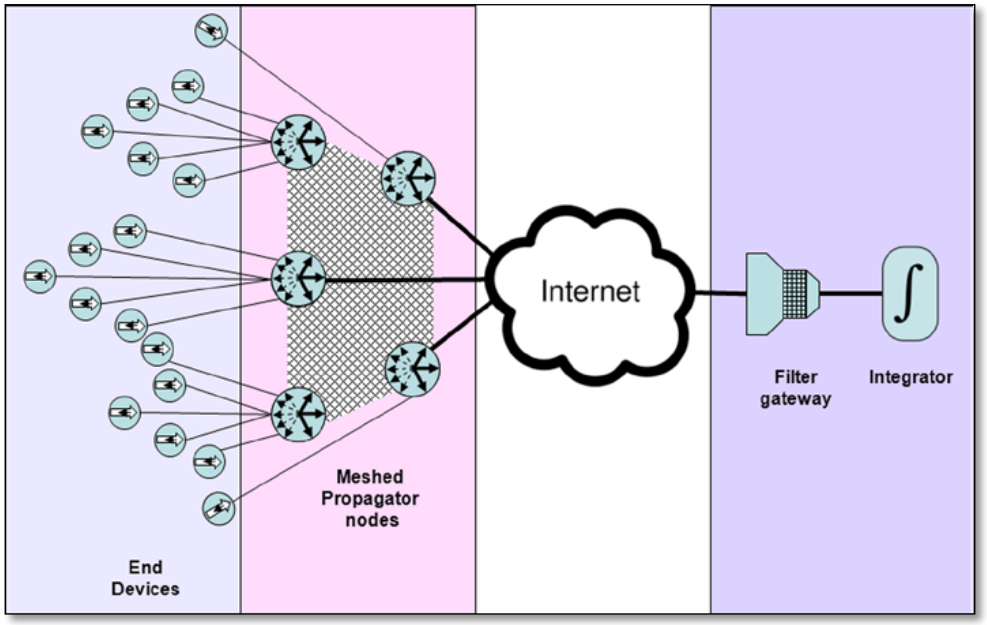


An Example: Smart Irrigation



(daCosta & Henderson, 2014)

A Simple Conceptual View



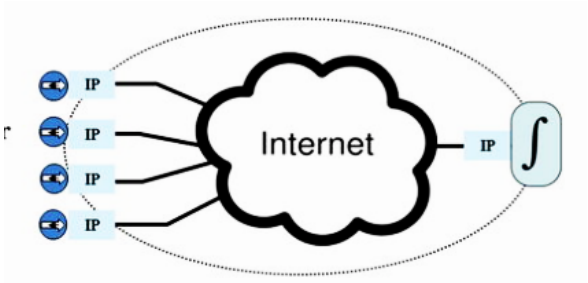
Speak & listen for data

Gateways to Internet (data flows)

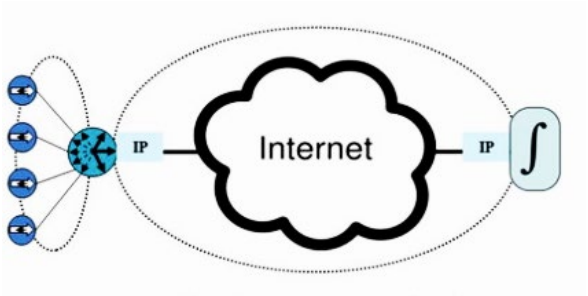
Collecting, integrating & acting (data analysis)

(daCosta & Henderson, 2014)

Addressing: Traditional vs. IoT



Traditional

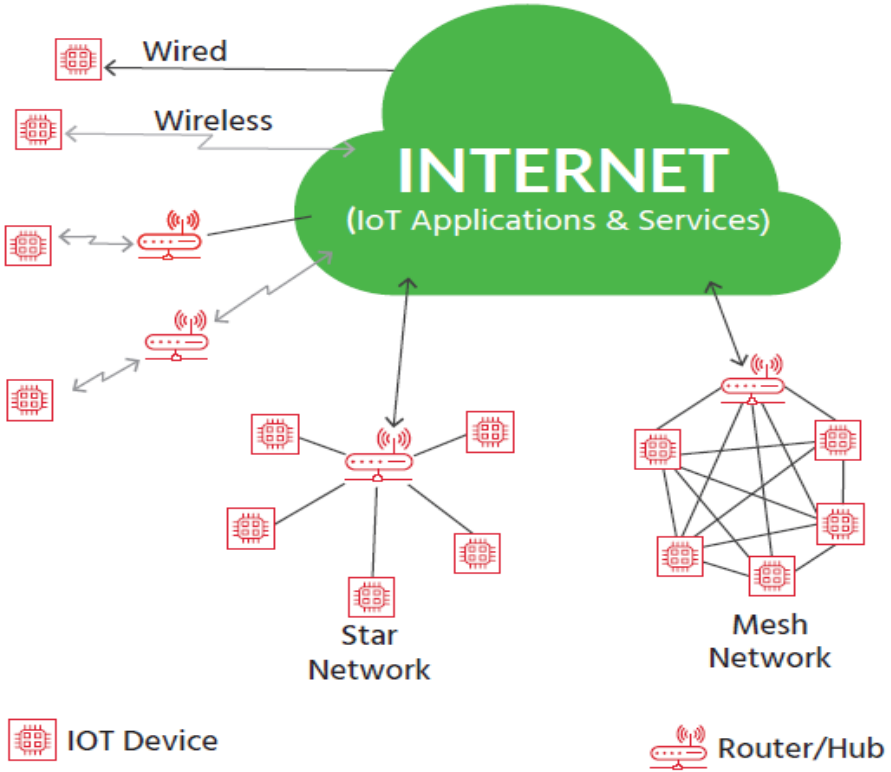


IoT

- Using IP communication protocols to transfer data from the sensor on the device through to some central repository.
- IPv4 → IPv6, Why?

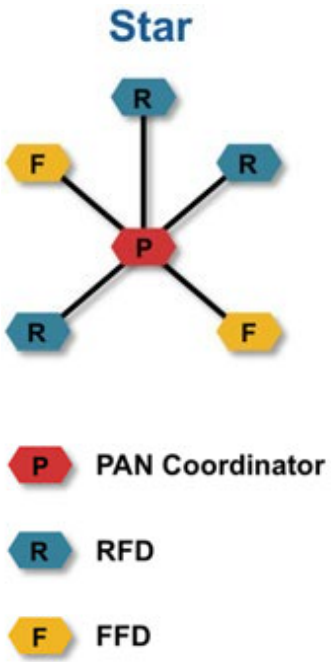
(daCosta & Henderson, 2014)

Connectivity to Internet



https://bridgera.com/internet-of-things/

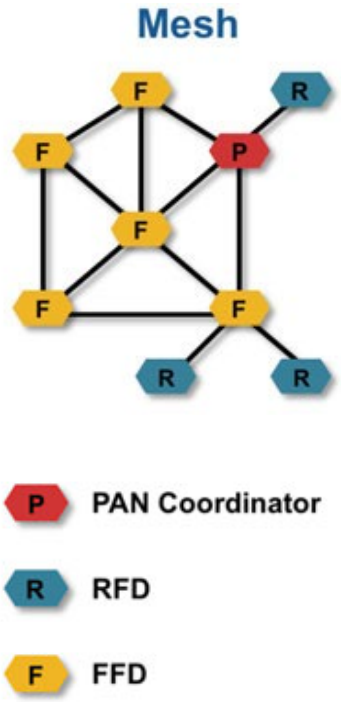
Star Topology



- All devices communicate through a single central controller (i.e. PAN coordinator).
- Hub-and-spoke model
  - Hub: PAN coordinator and powered
  - Spoke: other devices and most likely battery operated
  - FFD: Full-Function Device
  - RFD: Reduced-Function Device
- Each Star network operates independently of other networks
- Use cases
  - Smart homes
  - Personal health monitors

(Rayes and Salam, 2022)

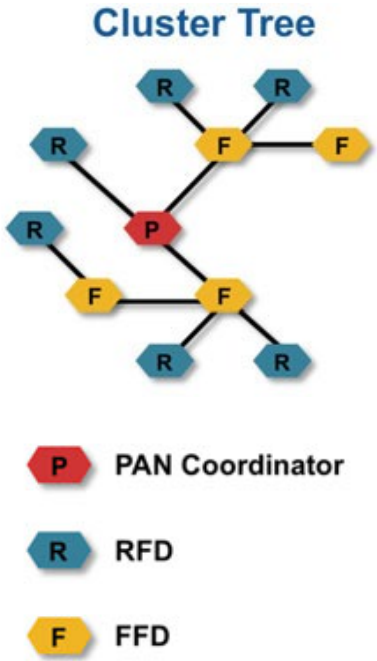
Mesh Topology



- Peer-to-peer
- Ad-hoc in formation
- Self-organising & self-recovering on node or link failures
- Multipath routing for reliability
- Use cases
  - Industrial control and process monitoring
  - Wireless sensor networks
  - Asset tracking
  - Precision agriculture
  - Inventory management

(Rayes and Salam, 2022)

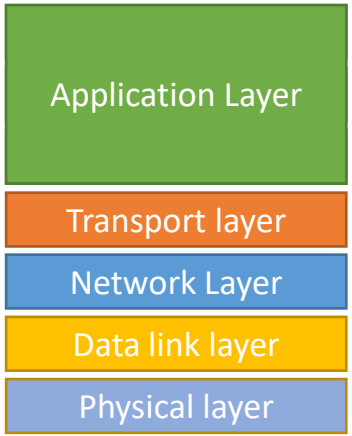
Cluster Tree Topology



- A special case of a mesh network with chained clusters
- Most of the devices are FFDs, and RFDs may connect as leaf node
- A single PAN coordinator forms the first cluster (as cluster head) with cluster identifier of zero, and each new device as a child device in its neighbour list
- To achieve large coverage area at the expense of increased message latency

(Rayes and Salam, 2022)

Internet Reference Model



Application interactions

Setup and maintain communication between devices

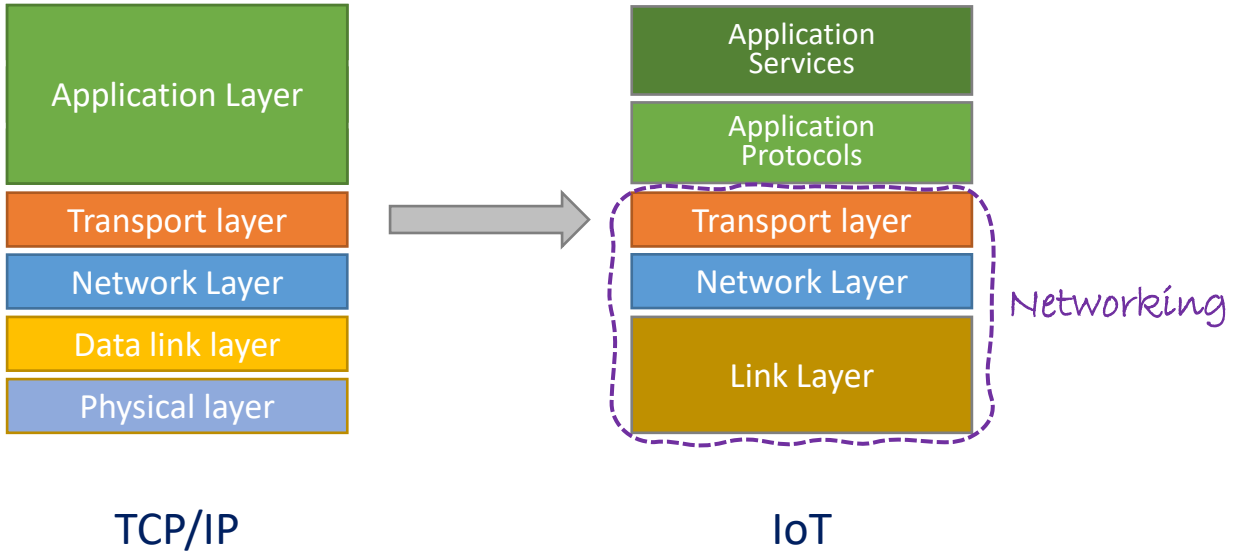
Global layer (IP addresses, routers)

Reliable communication link between IoT devices

Physical component

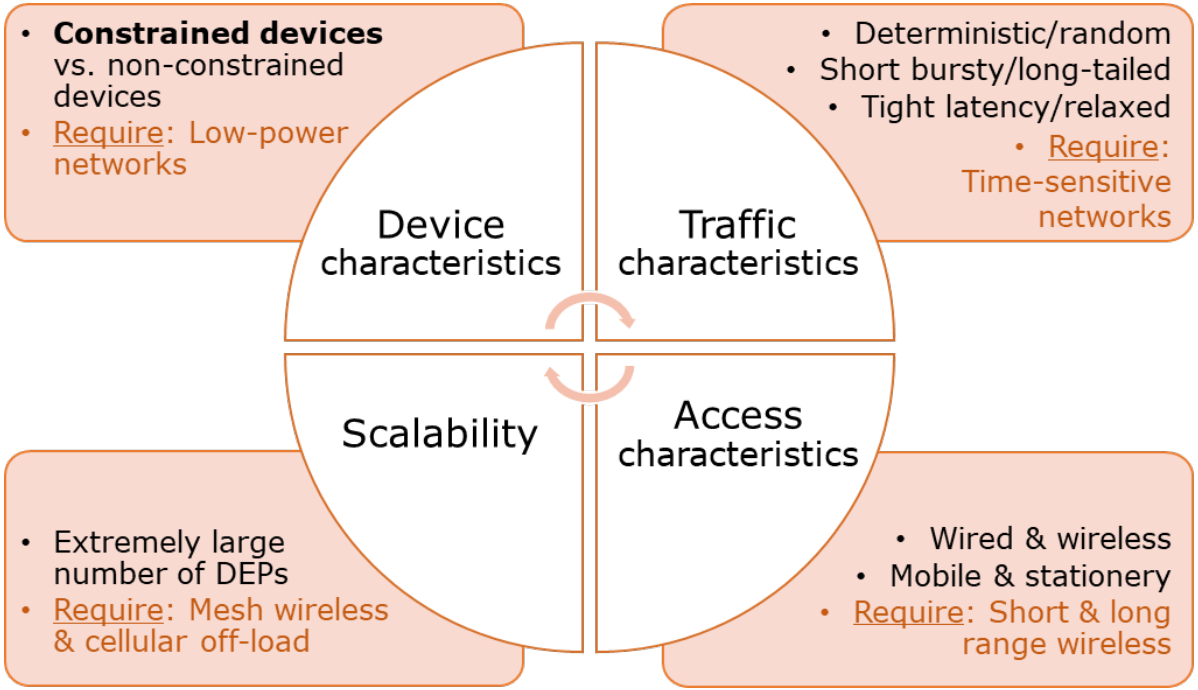
TCP/IP

IoT Protocol Stack



(Rayes and Salam, 2019)

Link Layer: Challenges



❖ Data Endpoint (DEP)

(Rayes and Salam, 2019)

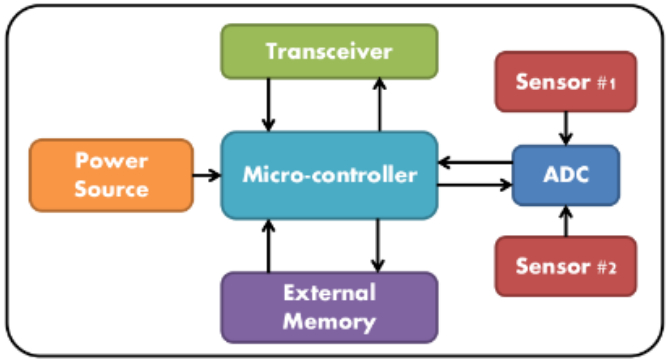
Constrained Devices

- **(Resource-)Constrained devices**: Small devices with limited processing, memory and power resources are referred to as constrained devices.
  - End nodes with sensors/actuators for a specific application purpose
  - Gateway-like devices connecting end nodes to low power lossy network and communicating with cloud/fog platforms
- Standard classes of constrained devices:
  - Class 0 (RAM: << 10 KB, ROM: << 100 KB)
  - Class 1 (RAM: ~10 KB, ROM: ~ 100 KB)
  - Class 2 (RAM: ~50 KB, ROM: ~250 KB)

(Rayes and Salam, 2019)

Support to Constrained Devices

- To address the use of constrained devices, **lightweight**, **energy efficient**, and **bandwidth-conscious** communication protocols are required across the entire IoT protocol stack.
- A common strategy for power-constrained devices is to apply sleep mode on the devices

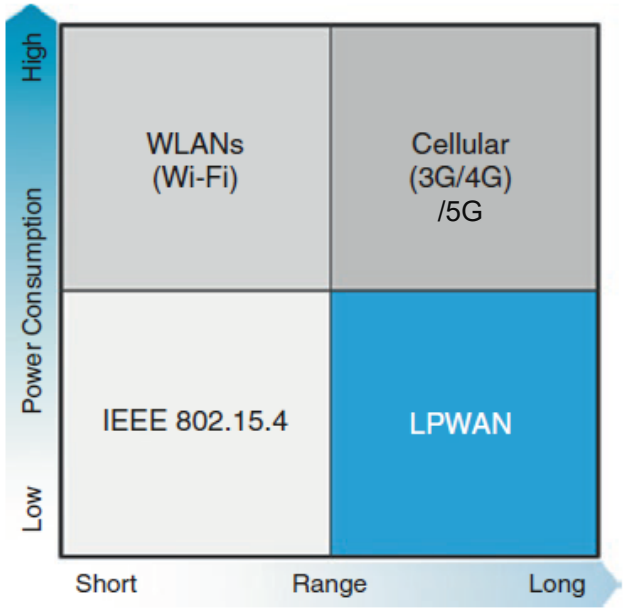


"Sensor Motes"

https://www.cisoplatform.com/profiles/blogs/classification-of-iot-devices

Wireless Communication Technologies

- Positioning of wireless communication technologies
  - Power Consumption
  - Network Range
- IEEE 802.15.4
- Low Power WAN (LPWAN)



(Rayes and Salam, 2022)

IEEE 802.15.4

- Packet-based radio protocol aimed at very **low-cost**, **battery-operated** devices
- The solution was meant to operate in an *unlicensed*, international frequency band to enable **short-range communication**.
- **Network topologies**
- Foundation for several protocol stacks including: Zigbee, 6LoWPAN, RPL, WirelessHART, Z-Wave, ISA 100.11a, etc.

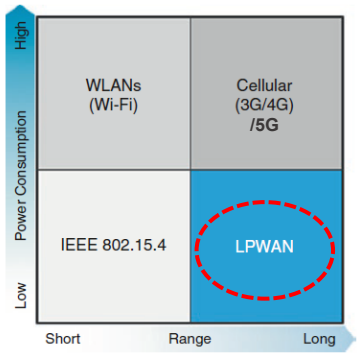
It supports two types of devices

- Full-function device (FFD)
  - Implements all the functions of the communication stack, which allows it to communicate with any other device in the network.
  - They can act as the PAN coordinator (in charge of its network domain).
- Reduced-function device (RFD)
  - For extremely simple devices with very modest resource and communication capabilities.
  - They can only communicate with FFDs and can never act as PAN coordinators.

(Rayes and Salam, 2022)

LoRaWAN for LPWAN

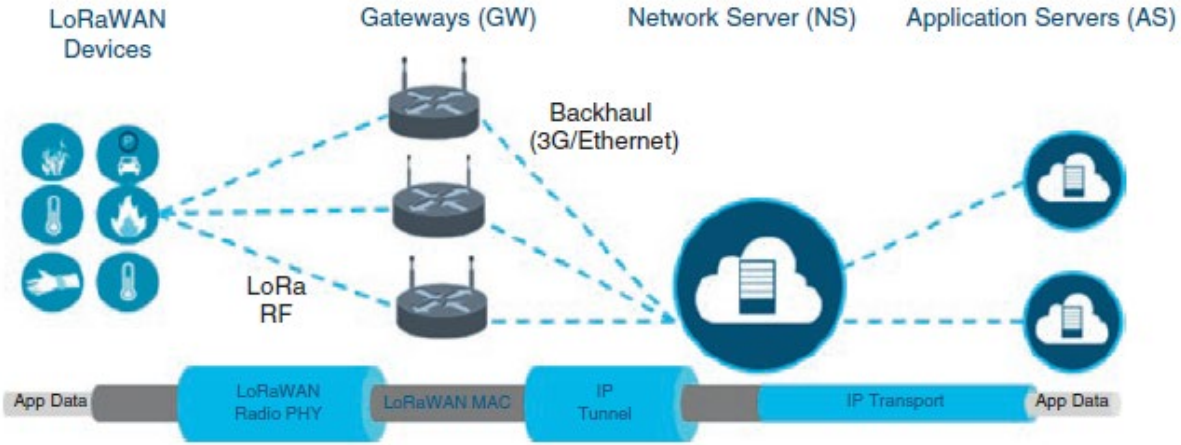
- LoRaWAN stands for Long Range Wide Area Network
- It defines a communication protocol and network architecture for LPWAN to address the requirements for **low power** consumption (i.e., long battery life), **long range**, and **high capacity** in LPWANs while maintaining low cost for the solution.
- LoRa: The communication protocol used in LoRaWAN
  - Extended range: a single base station can cover hundreds of square miles (enough to provide coverage over cities).



(Rayes and Salam, 2022)

LoRaWAN Network Architecture

- LoRaWAN employs a long-range *Star* topology
  - To minimise power consumption
  - Gateway acts as hub nodes, end devices form the spokes
  - No handover mechanism is required between gateways, thus supporting mobility of end nodes



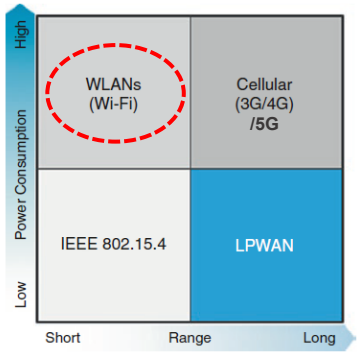
LoRaWAN end-to-end network architecture

(Rayes and Salam, 2022)

IEEE 802.11ah

- Provide extended range of Wi-Fi networks to support constrained devices for IoT
- Enhanced Wi-Fi technologies to address IoT requirements of low-power consumption and massive scalability
- Three main mechanisms for client stations
  - Power saving
  - Easier access to medium/channel,
  - Reduced overhead for communication

    (thus increasing throughput)

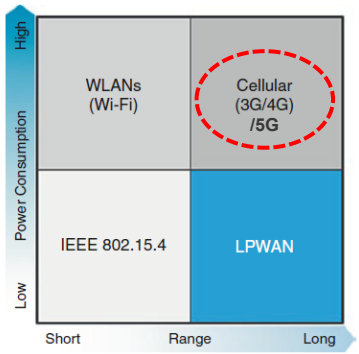


(Rayes and Salam, 2022)

LTE (Long Term Evolution)

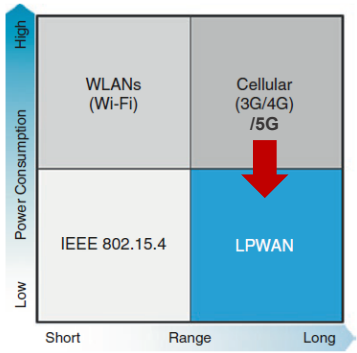
- LTE is a standard for wireless broadband communication for mobile devices and data terminals (e.g. 4G LTE)
- Targeting at high-speed cellular communications
- The highest data rate among the wireless technologies in use today
  - 1Gbps fixed speed and rate of 100 Mbps to mobile users
- LTE-M was proposed for M2M & IoT applications
- Use cases
  - Asset Tracking
  - Automated Meter Reading

https://en.wikipedia.org/wiki/LTE_(telecommunication)
https://en.wikipedia.org/wiki/LTE-M
https://www.digitaltrends.com/mobile/4g-vs-lte/



NB-IoT (Narrowband IoT)

- A LPWAN radio technology standard developed by 3GPP to enable a wide range of cellular devices and services for IoT
- Focus specifically on indoor coverage, low cost, long battery life, and high connection density
- Use a subset of the LTE standard, but limit the bandwidth to a single narrow-band of 200kHz
- Improved power consumption, system capacity, and spectrum efficiency.
- Use cases
  - Improved parking solutions
  - Smart rubbish bins
  - Street lamps



https://en.wikipedia.org/wiki/Narrowband_Io

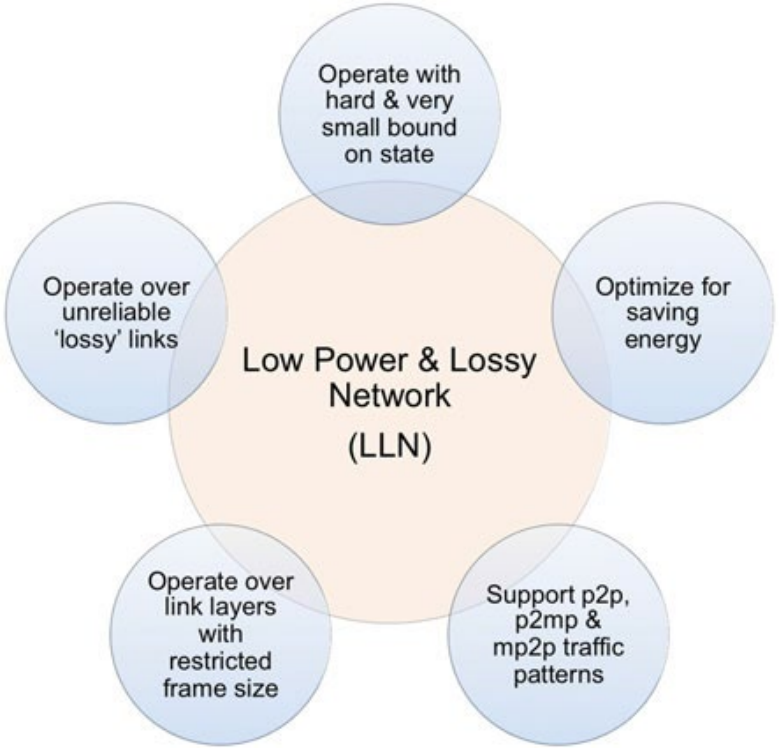
5G and IoT

- IoT is considered a key driver of 5G while IoT may also rely on 5G to reach its full potential
- IoT Requirements
  - 5G networks meet the standards set by organizations like the International Telecommunication Union (ITU) and the European Telecommunications Standards Institute (ETSI)
- Key use scenarios for 5G and IoT by ITU and ETSI
  - Enhanced mobile broadband (eMBB)
  - Massive/enhanced machine-type communications (mMTC/eMTC)
  - Ultra-reliable and low latency communications (URLLC)
- 5G & IoT
  - The migration from 4G to 5G to reach high speed, low power and low latency for massive IoT, tactile internet and robotics deployment

https://www.sdxcentral.com/5g/iot/definitions/5g-iot-standards/
https://www.iotforall.com/cellular-iot-explained-nb-iot-vs-lte-m/

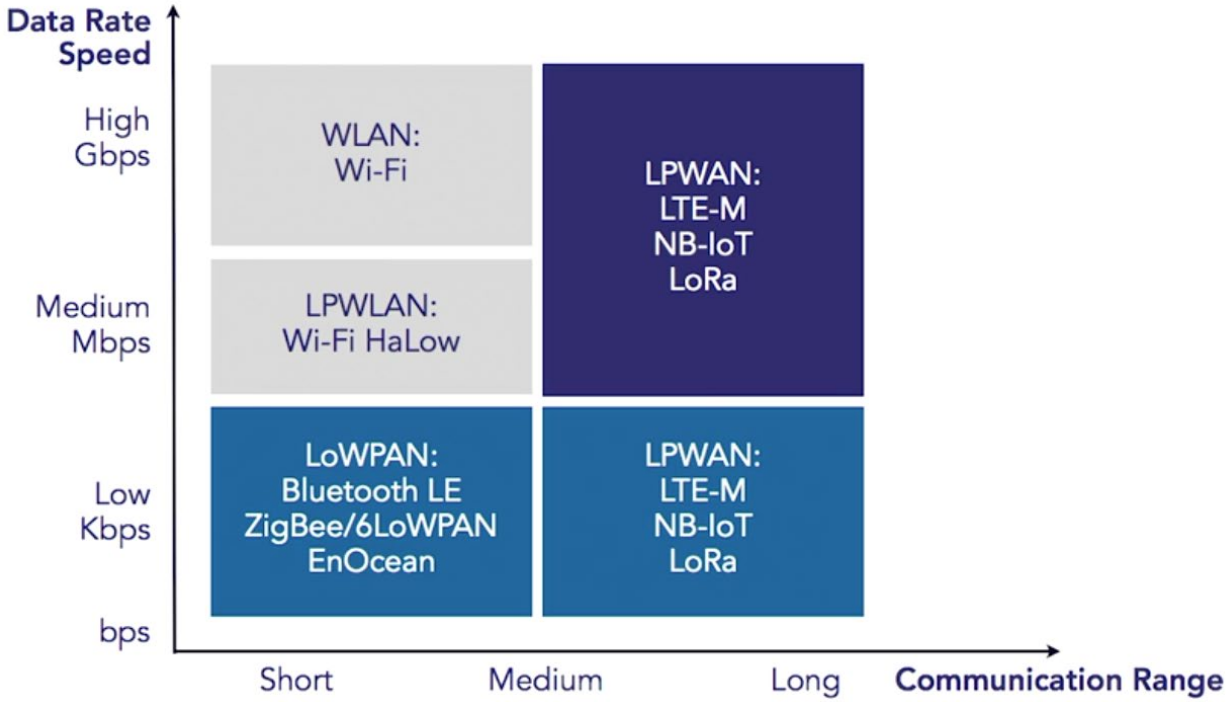
Network Layer: Challenges

- Most IoT deployments constitute low-power and lossy networks (LLNs).
- A wide scope of use cases of LLNs:
  - industrial monitoring,
  - building automation,
  - smart grid,
  - asset tracking
  - …
- LLN presents five challenges to the Network Layer of the IoT protocol stack.



(Rayes and Salam, 2019)

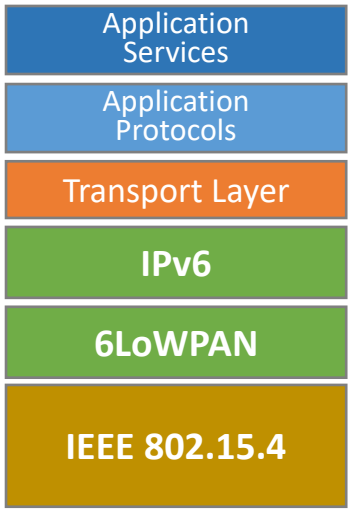
Wireless Technologies for LLN



(Vadada, 2017)

6LoWPAN

- 6LoWPAN stands for IPv6 over Low-power Wireless Personal Area Networks.
- Originally proposed as an adaptation layer for running IPv6 over IEEE 802.15.4 networks
- Often refer to any Link Layer **mesh network** built on low-power and lossy links using 6LoWPAN mechanisms
- Support very small devices with limited processing ability to transmit information wirelessly over Internet
- Use cases:
  - 6LowPan smart meters
  - Smart home (lighting, thermostats)



(Rayes and Salam, 2019)

6LoWPAN Functions

Adaption between IPv6 and IEEE 802.15.4

- Adapting the packet sizes
- Address resolution
- Different device designs
- Different focus on parameter optimization
- Adaptation for interoperability and packet formats
- Addressing management mechanisms
- Routing considerations and protocols for mesh topologies
  - **RPL**: IPv6 Routing Protocol for Low-Power and Lossy Networks
- Device and service discovery
- Security

https://en.wikipedia.org/wiki/6LoWPAN

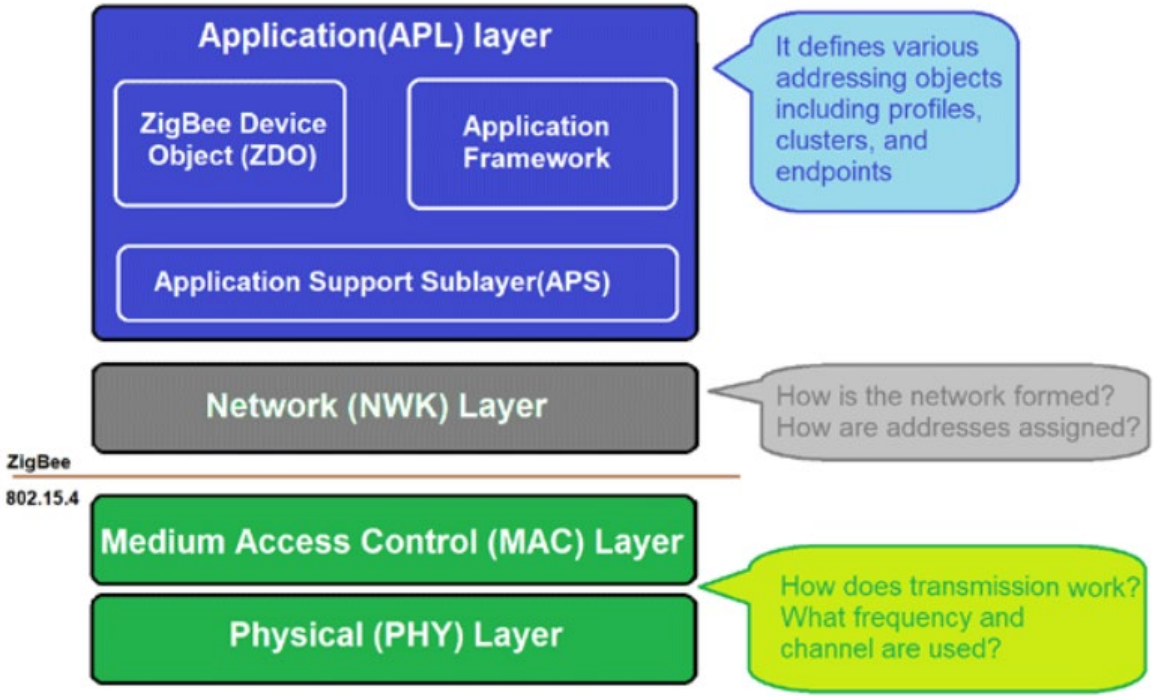
Zigbee

- Another specification for LoWPANs based on IEEE 802.15.4
- Highly popular, low-cost, low-power wireless mesh networking standard on the market
- Application: typically for personal or home-area networks, or in a wireless mesh for networks that operate over short ranges
- Use cases
  - Wireless Light Switches
  - Electrical Meters (Smart grid, demand response, etc)
  - Industrial Equipment Monitoring
  - ZigBee Range: A Tracking Device Use Case



https://en.wikipedia.org/wiki/Zigbee

A Zigbee Layered Architecture



https://circuitdigest.com/article/zigbee-introduction-architecture-at-commands

Zigbee Network

- Supports various network topology options, such as star, tree or mesh
- Several types of devices in a ZigBee network
  - End device: messaging, transmission and reception
  - Router: relaying messages between the nodes or end devices
  - Trust center: on a ZigBee router dedicated to security and network management
  - Coordinator: system initialization (e.g., start a network, or joining to a network)
- Note: *Z-Wave* is a wireless communications protocol used primarily for home automation. It is a mesh network using low-energy radio waves to communicate from appliance to appliance.

https://en.wikipedia.org/wiki/Zigbee

Technical Comparison:

IoT Network Technologies

Technology	Comm. Range	Data Rate	Latency	Power Consumption	Network Category
Ethernet	100 m (wired) per segment	> 1 Gbps	< 1 ms	High to medium	Wired Network
Wi-Fi	50 ~ 100 m	> 100 Mbps	< 10 ms	High	WLAN
4G (LTE)	5 ~ 20 km (area served by a single cellular-based station)	> 100 Mbps	30 ~ 50 ms	High	Cellular Network
IEEE 802.11ah (Wi-Fi HaLow)	1 km	150 kbps ~ 1 Mbps	30 ~ 100 ms	Low	LPWLAN
LTE-M	10 ~ 15 km	< 1 Mbps	10 ~ 100 ms	Low	LPWAN
IEEE 802.15.4	10 ~ 100 m	< 250 kbps	20 ms	Very Low	LoWPAN
ZigBee / 6LoWPAN	10 ~ 100 m	< 250 kbps	20 ~ 50 ms	Very Low	LoWPAN
NB-IoT	15 ~ 20 km	20 ~ 250 kbps	1.6 ~ 10s	Very Low	LPWAN
LoRa	15 ~ 20 km	0.3 ~ 50 kbps	Seconds	Ultra Low	LPWAN

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- Rayes, A. & Samer, S. (2019). Internet of Things – From Hype to Reality: The Road to Digitization (2nd edition). Springer.
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